

Chapter 1 – Introduction to Science

Hazard warning symbols & lab safety

Hazard warning symbols 危險警告標籤

Hazard Warning Symbol	Safety Precautions 安全預防措施	Examples in Daily Life and Lab
Flammable 易燃 	Keep away from heat and flame. 避免靠近熱源和火焰。	- Gasoline 汽油 (Daily) - Ethanol 乙醇 (Lab)
Corrosive 腐蝕性 	- Wear safety goggles (護目鏡) and protective gloves (保護手套) when handling the chemical. - Store in a well-ventilated (通風良好) room.	- Bleach 漂白水 (Daily) - Hydrochloric Acid 鹽酸 (Lab)
Explosive 爆炸性 	- Keep away from heat and flame. 避免靠近熱源和火焰。 - Avoid shock 震動, which may cause the chemical to explode.	- Fireworks 煙火 (Daily) - Ammonium Nitrate 硝酸銨 (Lab)
Toxic 有毒  Harmful 有害  Irritant 刺激性 	- Avoid breathing in the vapour 蒸氣 of the chemical. - Wear safety goggles and protective gloves when handling the chemical. - Wash your hands thoroughly 徹底洗手 after handling the chemical.	- Pesticides 殺蟲劑 (Daily) - Sodium Cyanide 氰化鈉 (Lab)

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Fill in blanks

1. Correction fluid 塗改液



What is this hazard warning label?

What precaution should we make when handling?

2. Bleach 漂白水



What is this hazard warning label?

What precaution should we make when handling?

3. Methanol 甲醇/工業酒精



What is this hazard warning label?

What precaution should we make when handling?

What is this hazard warning label?

What precaution should we make when handling?

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4. What hazard warning label can you find on the bottle of this chemical?



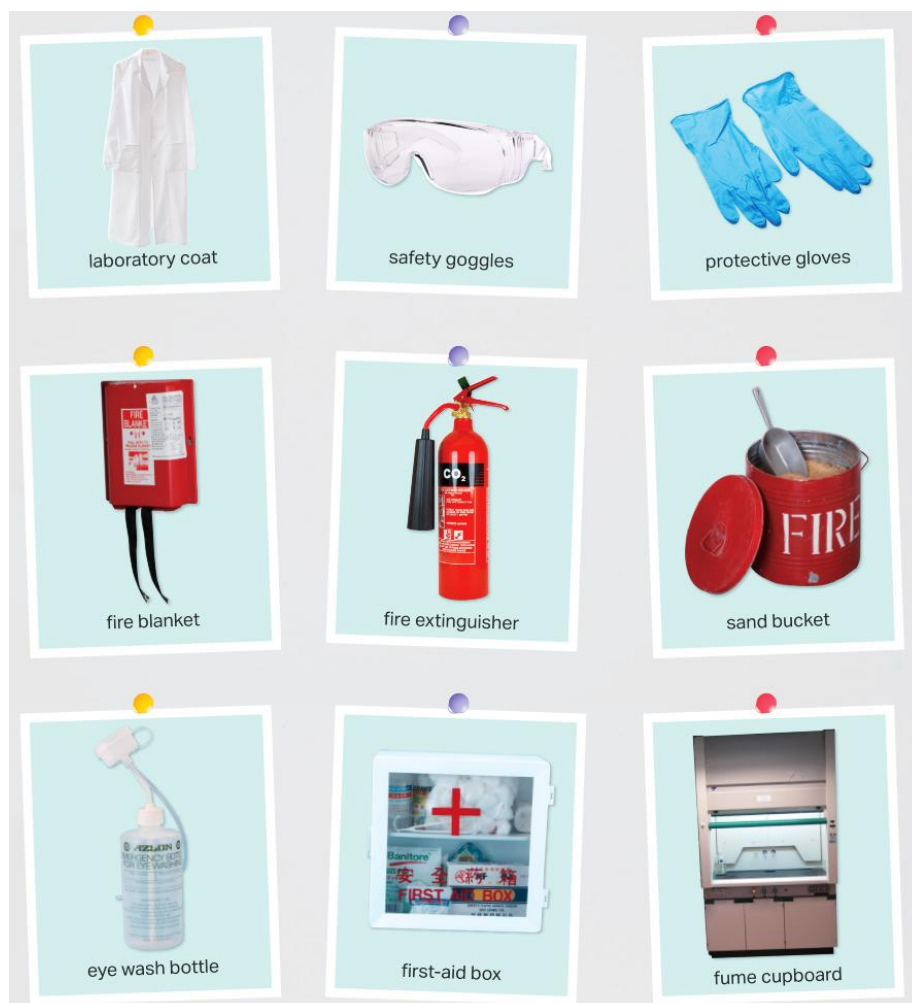
5. What hazard warning label can you find on the dynamite 炸藥?



6. What hazard warning label can you find on the bleach?



Lab safety equipment



Equipment Name	Primary Function
Laboratory Coat 實驗室外套	Protects the body and clothing from chemical spills and splashes 濺灑.
Safety Goggles 安全護目鏡	Protects the eyes from chemical splashes, fumes 煙霧, and flying debris 飛散物.
Protective Gloves 防護手套	Protects hands from harmful chemicals, heat, and biological contaminants 生物污染物.
Fire Blanket 滅火毯	Smothers small fires by cutting off the oxygen 氧氣 supply.
Fire Extinguisher 滅火器	Extinguishes fires by releasing fire-suppressing agents 滅火劑.
Sand Bucket 沙桶	Used to extinguish small fires.

Equipment Name	Primary Function
Eye Wash Bottle 洗眼瓶	Flushes 沖洗 and cleanses 清潔 the eyes in case of chemical exposure or irritation.
First-Aid Box 急救箱	Contains medical supplies 醫療用品 for treating minor injuries.
Fume Cupboard 通風櫥	Provides ventilation 通風 to safely work with hazardous 危險 chemicals and fumes 煙霧.

Accident 意外



Cuts
輕微割傷

Action 處理方法



Clean the wound with water and apply a dressing to it
用水清洗傷口，並用敷料覆蓋

Accident 意外



Heat burns or chemical burns
被熱物燙傷或被化學品灼傷

Action 處理方法



Place the affected area under slow running water
把受傷的部位放在流動緩慢的冷水下沖洗

Accident 意外



Chemicals get into the eyes
化學品濺入眼睛

Action 處理方法



Wash the eyes with distilled water from an eye wash bottle
用洗眼水瓶內的蒸餾水沖洗眼睛

Accident 意外



Glass breakage
打破玻璃儀器

Action 處理方法



Clean up and dispose of broken glass pieces in a broken glass container
用掃清理玻璃碎片，然後把碎片棄置在指定的容器內

Accident 意外



Gas leakage
氣體泄漏

Action 處理方法



Turn off the gas supply and open all windows and doors
立即關掉氣體開關掣及打開所有門窗

Accident 意外



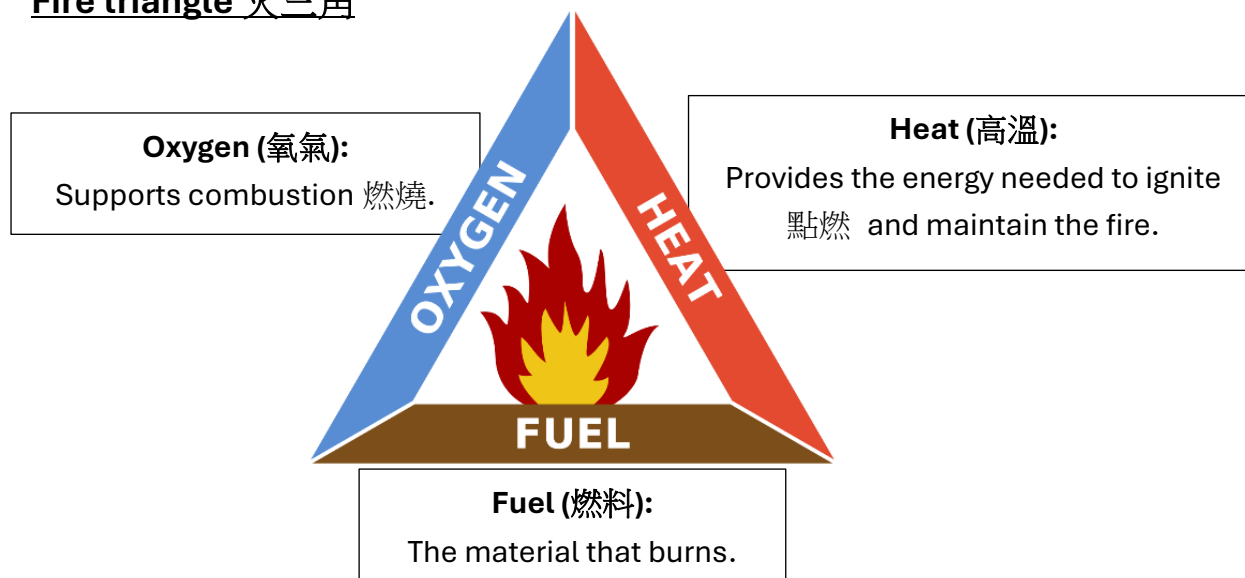
Fires
火警

Action 處理方法



Keep calm and report to your teacher at once
保持鎮定並立刻向老師報告

Fire triangle 火三角



• extinguish fire 滅火

Method		Main Action Against Fire Triangle
Fire Extinguisher 滅火器		→ Removes heat → Remove oxygen (e.g. CO ₂ 二氧化碳)
Fire Blanket 滅火毯		→ Removes oxygen
Sand Bucket 沙桶		→ Removes oxygen
Water		→ Removes heat → Remove oxygen
Fire Break 防火帶		→ Removes fuel by creating a gap 間隔 between the fire and flammable materials, e.g. cutting of trees

Spelling exercises

Section 1: Equipment Identification

Complete the spelling for described safety equipment:

1. Eye protection against chemical splashes: S _____ G _____
2. Container using sand to smother flames: S _____ B _____
3. Ventilation device for chemical work: F _____ C _____
4. Emergency eye washing device: E _____ W _____ B _____
5. Oxygen-blocking fire suppression fabric: F _____ B _____

Section 2: Emergency Protocols

Fill in missing words for accident responses:

1. Broken glass handling: Dispose in _____ G _____ container
2. Eye contamination: Flush with _____ water for 15 mins
3. Gas leak procedure: First shut off G _____ supply
4. Burn treatment: Use S _____ running water on affected area

Section 3: Fire Triangle

Complete fire safety concepts:

1. Fire blankets target the _____ element in fire triangle
2. Water primarily removes _____ from fires
3. Fire breaks eliminate _____ from combustion
4. CO₂ extinguishers work by displacing O _____

Answer Key

Hazard warning symbols 危險警告標籤

1. Flammable; Keep away from heat and flame
2. Corrosive ; Wear safety goggles and protective gloves when handling the chemical/ Store in a well-ventilated room
3. Flammable; Keep away from heat and flame

Toxic; Avoid breathing in the vapour of the chemical
4. Corrosive, Flammable, Toxic
5. Explosive
6. Irritant

Spelling exercises

Section 1: Equipment Identification

1. Safety Goggles
2. Sand Bucket
3. Fume Cupboard (or Fume Hood)
4. Eye Wash Bottle
5. Fire Blanket

Section 2: Emergency Protocols

1. Broken Glass
2. Distilled
3. Gas
4. Slow

Section 3: Fire Science

1. Oxygen
2. Heat
3. Fuel
4. Oxygen